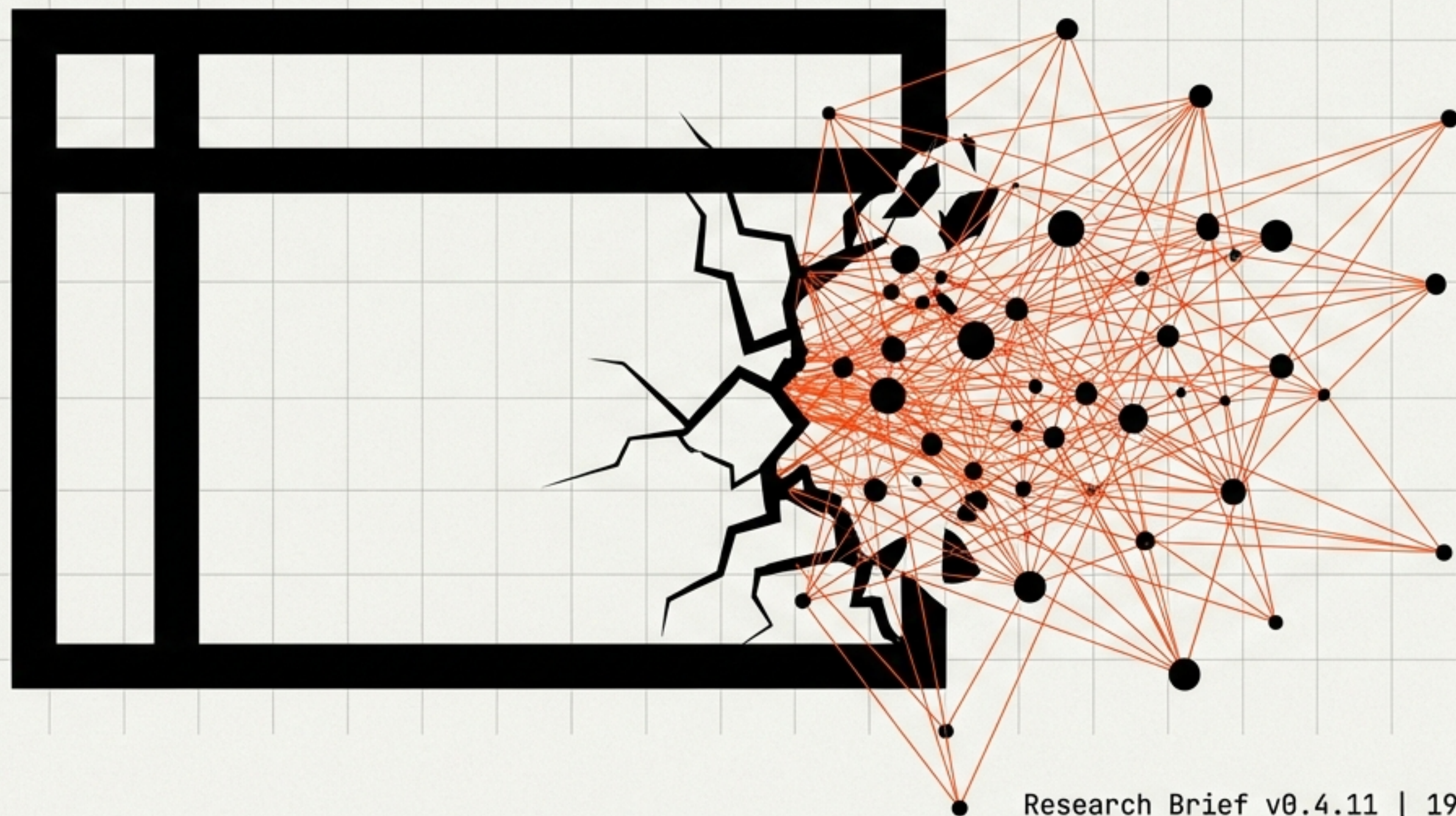


Tables as Graphs

Architectural Philosophy & The Nature of Data Storage



This Is Not a Metaphor

Everything can be represented as a graph. This is a statement about the nature of data, not a choice of technology.

- A CSV file is a graph.
- A Postgres database is a graph.
- A text file containing 'Hello World' is a graph.

TEXT FILE



DATABASE TABLE

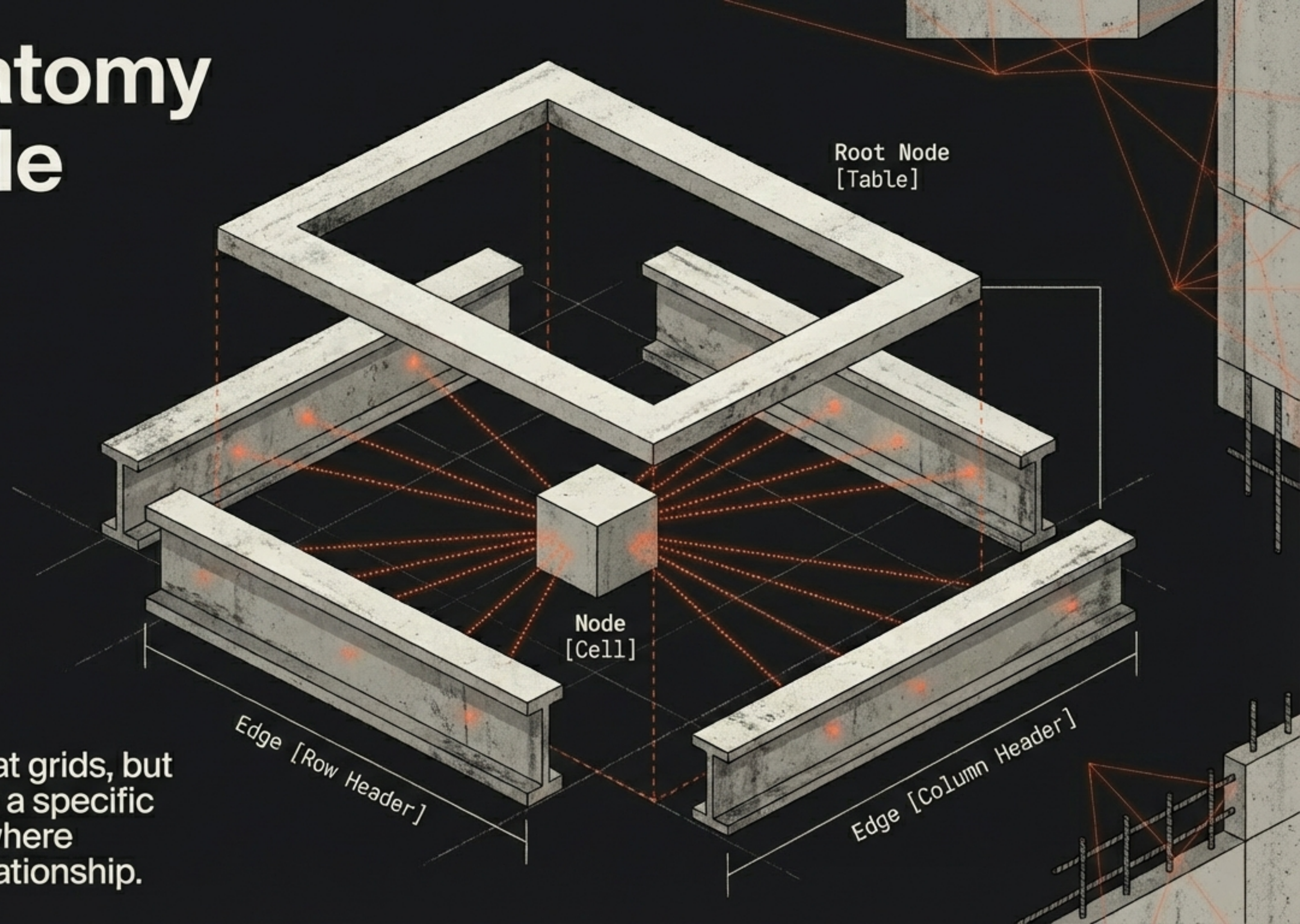
ID	NAME	AGE	OCCUPATION	CITY
001	Alice	30	Engineer	Berlin
002	Bob	25	Designer	London
003	Charlie	35	Manager	New York

SPREADSHEET

	A	B	C	D
1				
2	Date	Product	Sales	Region
2	2024-01-01	Widget A	\$1000	North
3	2024-01-02	Widget B	\$1500	South
4	2024-01-03	Widget C	\$2000	East

The question is never “is this data a graph?”—it always is. The question is: **What is the most optimal way to store, retrieve, edit, visualise, and version control this graph?**

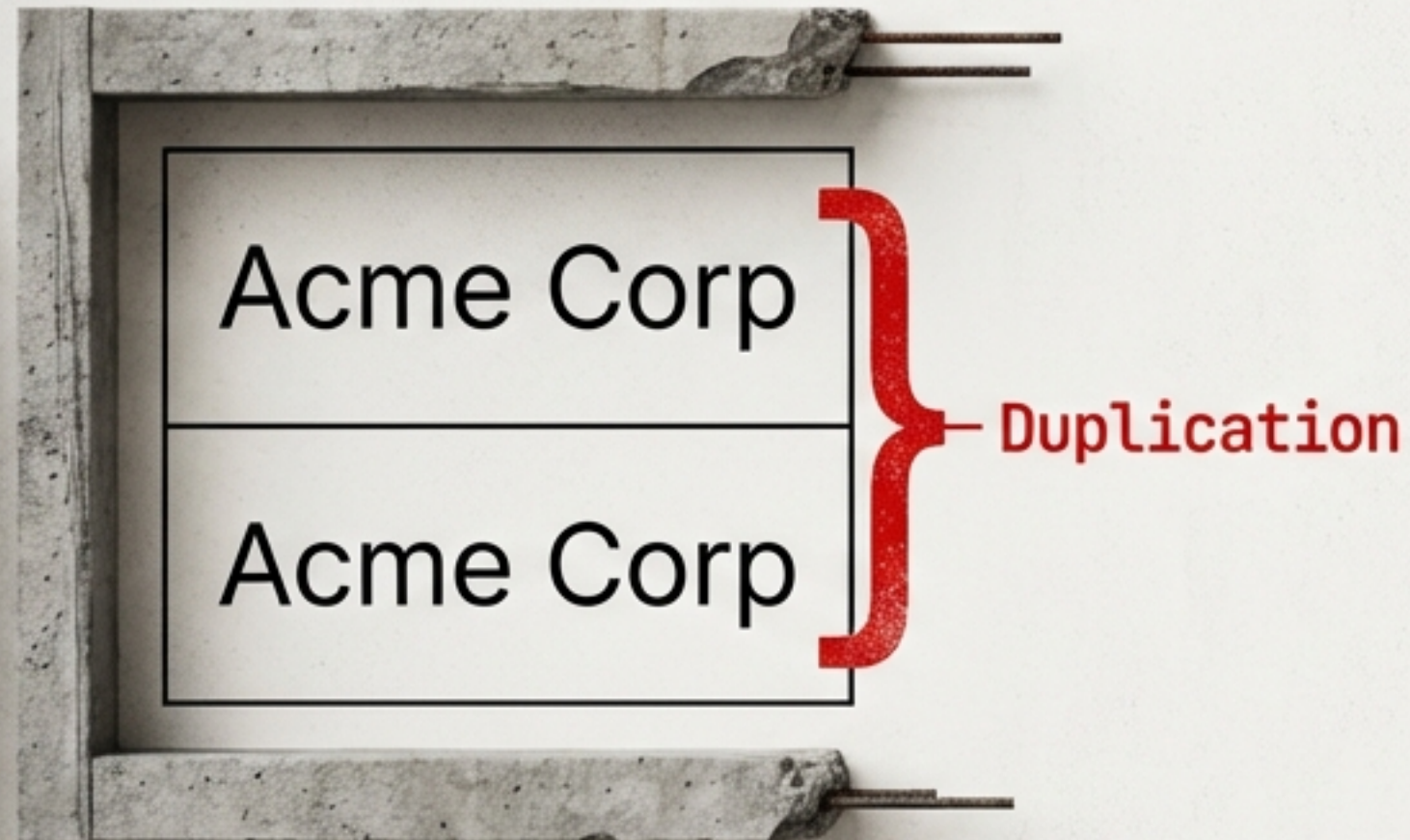
The Anatomy of a Table



We treat tables as flat grids, but structurally, they are a specific type of rigid graph where position dictates relationship.

The Non-Unique Node Problem

Tables duplicate data to display it.



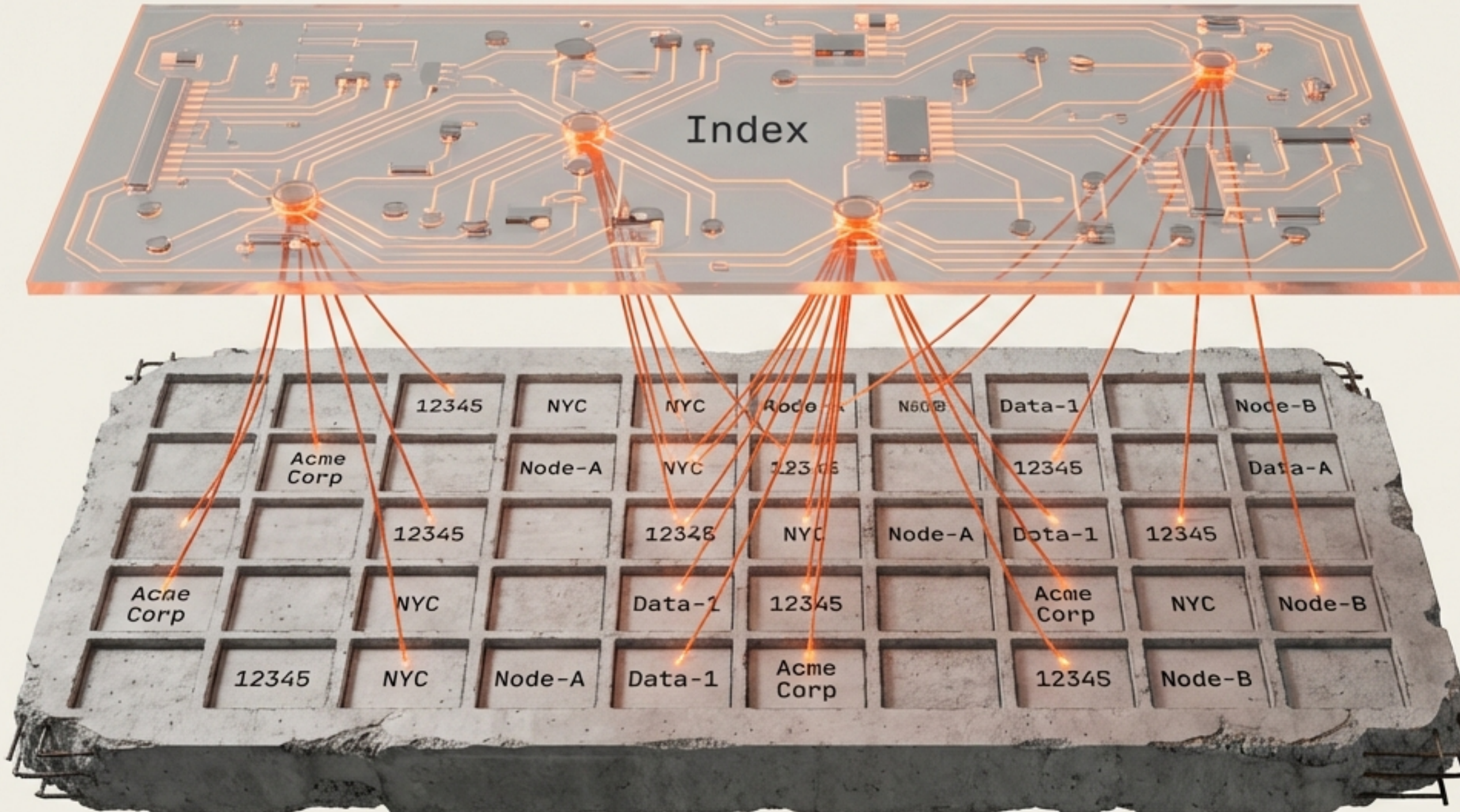
Native graphs deduplicate data to store it.



**In a table, relationships are implied by repetition.
In a graph, relationships are explicit connections.**

An Index is a Table Trying to be a Graph

Neue Haas Grotesk Display



A table is a graph optimised for retrieval, not for relationships. When we need relationship capabilities in a database, we build an **Index**.

An index is a subgraph that creates unique nodes for repeated values and links them back to the cells where they appear.

It retroactively adds the connectivity that a native graph possesses by default.

The Pragmatic Question

If everything is a graph, why not store everything in a Native Graph DB?



Human
Readability &
Editability



Traversal Speed
& Connectivity

Because the storage medium depends on how the human needs to interact with the data. The important thing is not that everything is **stored** in a graph database. The important thing is that everything can be **understood** as a graph and **connected** into one.

The Storage Spectrum

Storage Medium	Optimised For	Trade-Offs
Native Graph DB (Neo4j)	Relationship traversal, multi-hop queries.	Worse at bulk retrieval; less familiar to users.
Relational DB (SQL)	Indexed retrieval, structured queries.	Schema-heavy; worse at deep traversal.
Flat Files (.issues, .md)	Human readability, Git version control.	No query engine.
Spreadsheets (Excel)	Visual editing, collaboration.	Poor version control; limited programmability.

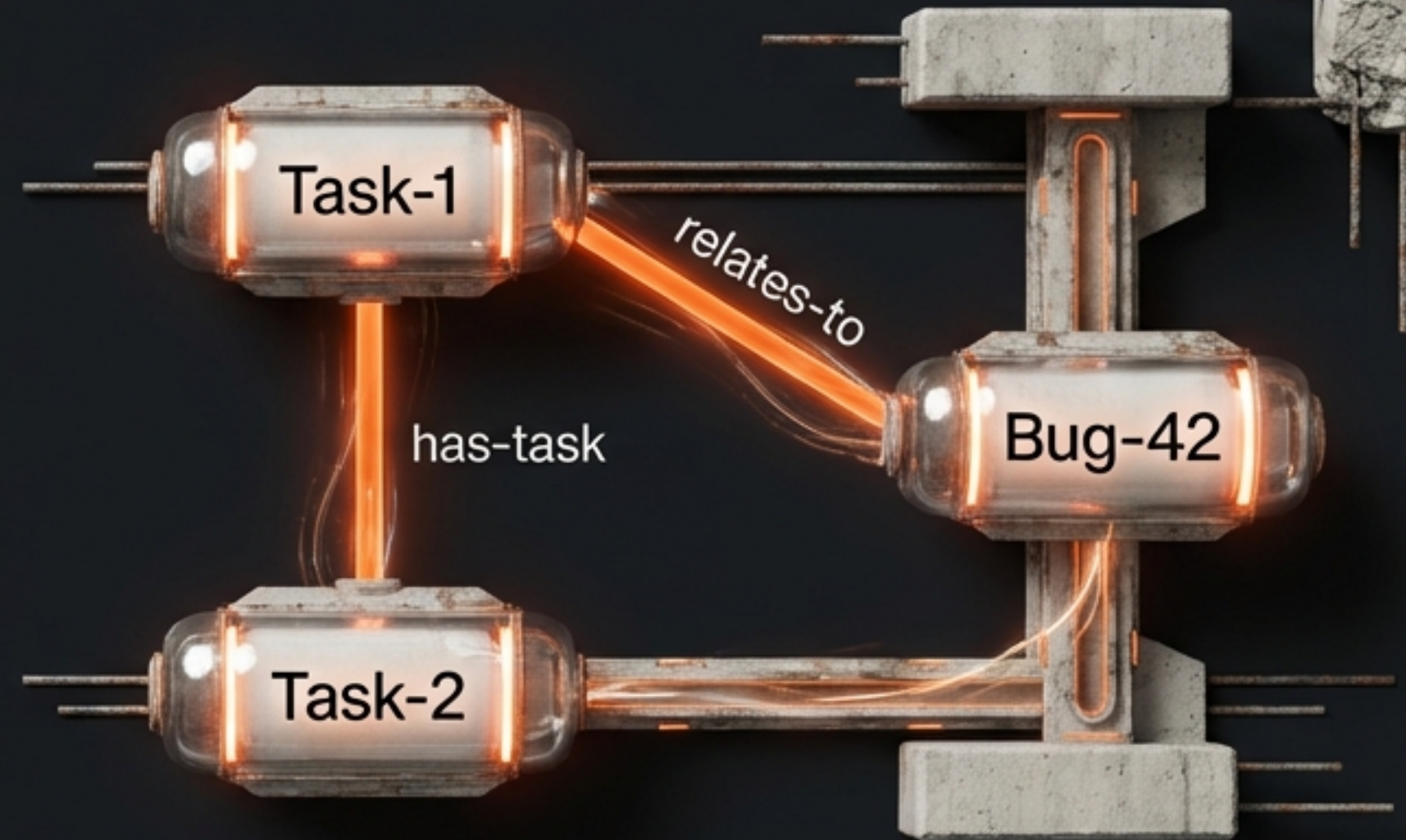
Takeaway: Use what exists. Ensure we can link into it.



Proof of Concept: Issues-FS

A format that looks like a table but acts like a graph.

```
[Task-1] | Todo | Fix login bug  
-> [Bug-42]  
[Task-2] | Done | Write tests
```



The user operates in table mode. The **system stores and connects** in graph mode.

Extension Path: CSV & XLSX

	A	B	C	D	E	F
1	ID	Title	Status	Parent		
2	T-1	Login Bug	Open	-		
3	T-2	Write Tests	Done	T-1		
4	B-1	API Error	Open	T-1		
5						
6						
7						
8						
9						
10						
11						

Parent: T-1

T-1

ID: T-1

Title: Login Bug

Status: Open

T-2

ID: T-2

Title: Write Tests

Status: Done

B-1

ID: B-1

Title: API Error

Status: Open

Mapping Rules:

1. Row → Node
2. Column Header → Property
3. "Parent" / "Relates-To" Column → Edge

XLSX Specifics:

Each sheet represents a different node type or projection. Cross-sheet references become cross-graph edges.

Live Collaboration



Google Sheets

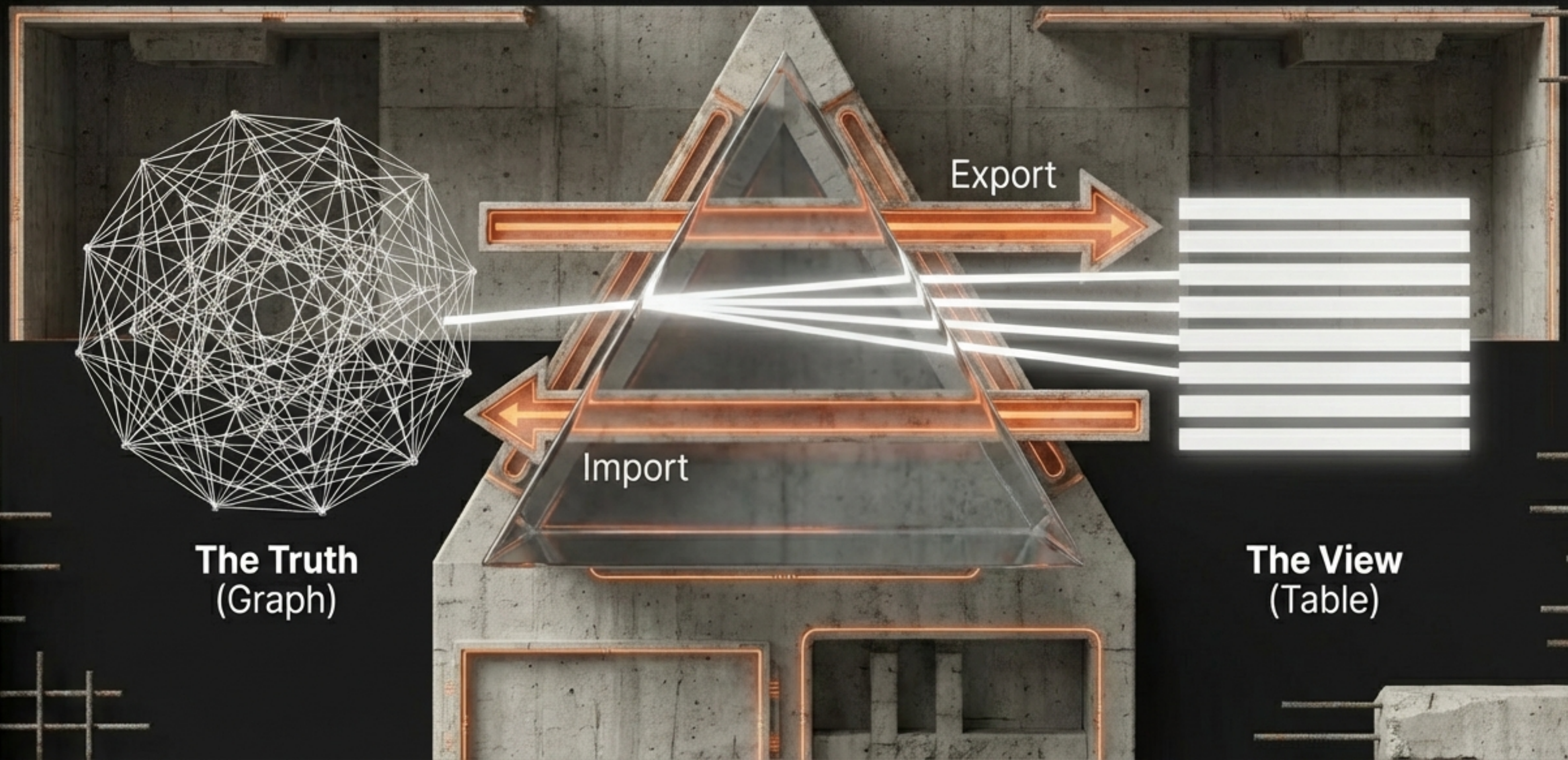
Google Sheets: Bi-directional sync via API.
Edit a cell → Graph node updates.
Graph writes back → Sheet status updates.

Airtable

Airtable: Native 'Linked Records' are explicit edges.
Table → Node Type.
Linked Field → Edge.

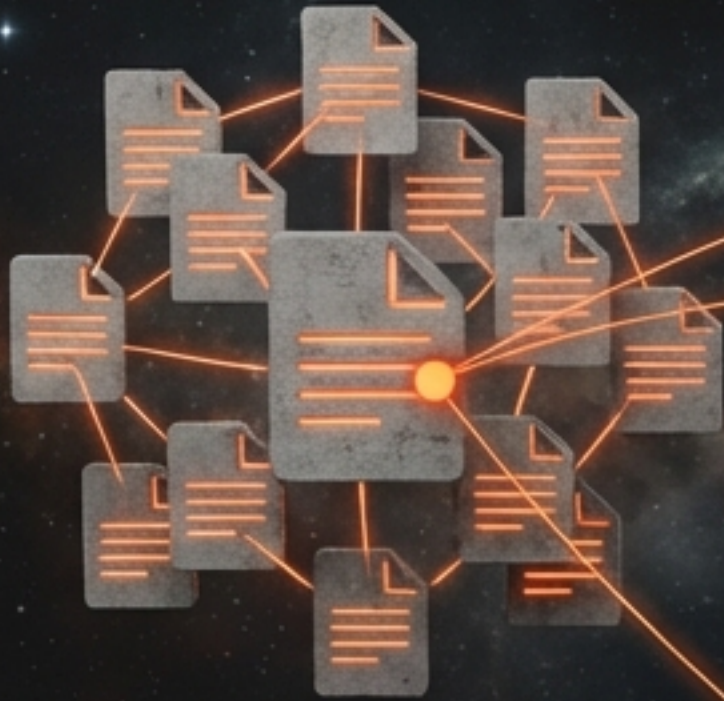
Projections: The Bridge

Tables are for presentation. Graphs are for storage. The Projection is the translation layer.



Graphs of Graphs

The Fractal Philosophy



Flat Files



Spreadsheets



Postgres DB

Each system stores its local graph in the optimal medium.

- A node in Issues-FS links to...
- A row in Google Sheets which links to...
- A record in Postgres.

Optimisation Dimensions

Dimension	Database	Spreadsheet	Flat Files (.issues)
Store	Structure	Flexibility	Portability
Retrieve	Lookup	Visual Scan	Simple Read
Edit	SQL	UI Click	Text Editor (Fast)
Version Control	Logs	None	Git (Perfect)

.issues is chosen because it optimises for **Edit**, Version Control, and Human **Readability**, while remaining a graph underneath.

The User Operates in Table Mode. The System Stores in Graph Mode.

We don't ask users to change their tools.
We link into what they already use.